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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – ECE / CSE & B.TECH - IT

20EC211 - EMBEDDED SYSTEMS AND INTERNET OF THINGS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1 Identify the stage that combines multiple object files into a single executable file?**
a) Compilation b) Linking c) Locating d) Debugging
2. **CO2 char *msg = "System Ready"; Which code should replace it for MISRA compliance?**
a) char msg[] = "System Ready"; b) const char *msg = "System Ready";
c) char * const msg = "System Ready"; d) char msg[20];
3. **CO3 The root file system in Embedded Linux contains _____.**
a) Kernel source code b) Configuration files and device drivers
c) Libraries, utilities, and user applications d) Hardware registers
4. **CO4 If the clock tick rate is too high _____.**
a) The system becomes more efficient b) Context switching stops
c) Tasks execute faster d) CPU time is wasted on frequent interrupts
5. **CO5 Identify the communication component in the physical design of IoT.**
a) Cloud platform b) RFID c) Database d) Application Software
6. **CO5 SaaS (Software as a Service) provides users with _____.**
a) Only hardware resources b) Ready-to-use software applications over the internet
c) Development tools to build applications d) Operating systems only

7. **CO3** Which of the following is an example of a character device?

- a) Hard disk
- b) Keyboard
- c) Network interface card
- d) USB mass storage device

Fill in the blanks:

(3x 1 = 3 Marks)

- 8. **CO1** An _____ is a hardware or software tool that mimics the behavior of a real microprocessor or microcontroller to allow testing and debugging.
- 9. **CO3** _____ is responsible for initializing hardware and loading the Linux kernel
- 10. **CO4** _____ is the process of saving the state of the current task and loading the state of the next scheduled task.

PART - B

(5x 3 = 15 Marks)

- 11. **CO1** Illustrate an embedded system and also list any four constraints typically associated with embedded systems.
- 12. **CO2** Write the compliance and non-compliance code using MISRA C for the rule “In an enumerator list, the “=” construct shall not be used to explicitly initialize members other than the first, unless all items are explicitly initialized.”
- 13. **CO3** Define a memory leak and outline the working of any two Linux tools used to detect memory leaks.
- 14. **CO4** State the differences between priority inheritance and priority inversion.
- 15. **CO5** A national bank is planning to adopt cloud computing for its operations. Identify the most suitable cloud deployment model and justify your choice.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

- 16. **CO2** i) Explain any 4 rules of MISRA-C with its compliance and non-compliance code. (10)
- CO1** ii) Draw the block diagram of a digital camera's embedded system and explain how the characteristics of being 'single-functioned,' 'tightly constrained,' 'reactive,' and 'real-time' apply to the system. (5)

17. **CO1** i) Explain the significance of memory testing in embedded systems. Identify the strategies employed to ensure memory reliability in critical applications. (10)
- CO1** ii) A company is designing a smart wearable device. Should they use a general-purpose processor or an application-specific one? Justify your recommendation based on cost, performance, and power efficiency. (5)
18. **CO3** i) Explain the architecture of the Linux I2C subsystem and discuss the communication between CPU and I2C devices through adapters. (10)
- CO3** ii) Discuss the importance of GNU Binutils in software development and illustrate the purpose of any five commands with suitable examples. (5)
19. **CO4** i) Using the Round Robin (preemptive) CPU scheduling algorithm with a time quantum of 2 ms, schedule three processes—P1, P2, and P3—that arrive simultaneously at time 0 with burst times of 10 ms, 5 ms, and 8 ms respectively. Apply the algorithm to draw the Gantt chart and compute the average waiting time and average turnaround time for all processes (10)

Process	Arrival time	Burst time
P1	0 ms	10 ms
P2	0 ms	5 ms
P3	0 ms	8 ms

- CO4** ii) In RTOS assume Task 1 and Task 2 holds the resources unnecessarily leads to indefinite waiting. Identify the reason for the occurrence and suggest ways to prevent this condition. (5)

20. **CO4** i) Use the Shortest Job First (SJF) non-preemptive CPU scheduling algorithm for the processes P1, P2, P3, and P4 with the given arrival times, burst times, and priorities to construct the Gantt chart, determine the average waiting time, and find the average turnaround time for all processes. (10)

Process	Arrival time	Burst time	Priority
P1	0 ms	5 ms	1
P2	1 ms	3 ms	2
P3	2 ms	8 ms	1
P4	3 ms	6 ms	3

- CO4 ii)** Define a reentrant function. List its features and give examples of reentrant and non-reentrant functions. (5)
21. **CO6 i)** Design a system that integrates motion detection into a disaster management framework. Give a functional block diagram and a sample embedded C program to detect motion and communicate the same with the disaster management network. (10)
- CO5 ii)** List the differences between IAAS, PAAS and SAAS. (5)
22. **CO5 i)** Illustrate the physical architecture of an IoT system using a block diagram and explain the function of each component in the system. (10)
- CO6 ii)** Explain the mechanism of Air Quality Monitoring System to notify users about dangerous pollution levels. (5)
